

Presentation Script

2.

- The experiment setup is a 10-armed bandit, each with randomised expected reward with Gaussian distributions.
- With seed 5, the optimal arm with the highest expected reward is arm index 2.
- The simulation settings for each question is summarised in the table

3.

- When a suboptimal arm is selected in an action, reward is lost relative to the optimal arm.
- Regret is the cumulative expected regret up to time T .
- Mathematically, sublinear regret means the limit as T goes to infinity, the regret over T is 0.
- It means that the regret grows slower than linearly in T , so the average regret per time step vanishes.
- It is important because it indicates that the average reward approaches the optimal expected reward; it also means that we can safely do exploration without concerning not able to reach optimal.
- In the workshop, we estimated the reward by averaging pseudo-regret over the M runs.

4.

- For our experiment setup there are a few issues for testing sublinearity of epsilon greedy algorithms.
 - Sublinearity is an asymptotic property
 - Simulation only gives finite horizon
- To handle this, we used three diagnostics together:
 1. The normalised regret, which by definition of sublinear regret, should approach 0 as T grows.
 2. The incremental slope, measuring the marginal regret growth, which should be decreasing for sublinear regret
 3. Effective exponent, estimating the local growth order, which should be below 1 for sublinear regret

5.

- Just a brief explanation of the effective exponent:
 - What we do here is to model $R_{\hat{}}$ locally as a power law.
 - Then we can obtain an approximation of this exponent in terms of $R_{\hat{}}$ and T
 - And it is obvious to see that for sublinear regret it is necessary for the exponent to be strictly smaller than 1.
 - whereas if it's 1 or approaching 1 it could be a signal for linear regret.

6.

- The numerical algorithm utilising these would be:
 - For each horizon T_i , run the policy M times, run the epsilon algorithm, compute the pseudo regret $R_{\hat{}}$ for each T_i .
 - Then compute the incremental slopes and effective exponents using $R_{\hat{}}$.
 - Finally examine them based on the mentioned criteria.

7.

- Here are the plots for the three diagnostics
 - $\epsilon = 0$, $\epsilon = 0.10$ and 0.90^t show clear non-sublinear behaviour as their exponent tend toward 1
 - The polynomial schedule is the only one with all three diagnostics supporting sublinear regret.
 - $\epsilon = 0.01$ and 0.99^t are numerically inconclusive over our this finite horizon, because their effective exponents are below 1 but still show trend of going towards 1.

8.

- The interpretation for the algorithms that can be concluded would be:
 - Fixed ϵ explores forever and therefore incurs linear regret asymptotically.
 - Polynomial decay reduced exploration gradually while maintaining sufficient exploration.
 - whereas exponential decay removes exploration too quickly and lock-in suboptimal arm.